

Self-assessment questions on Lecture n.10

- Explain what are the fundamental requirements for a passivating film to protect metals from corrosion
- Clear up why H contributes to augment corrosion and in which circumstances it is likely to be crucial
- Explain the role of Al, Cu, Cr, Ti, as metals as well as alloying elements, on corrosion
- Elucidate how the area factor can be used to contrast corrosion and which type of corrosion mechanism
- Clarify the significance of the electrode potential Table, and its potential use to assess the corrosion rate
- Elucidate the factors affecting galvanic/pitting/uniform/interstitial corrosion
- Illustrate the factors affecting pitting corrosion
- Explain how rust is formed in corrosion of steels and its chemical composition
- Indicate three design/materials-selection factors which may promote corrosion
- Explain the role of Cr, Ni and Mo in stainless steels and in which steels are all important
- Rank the corrosion resistance of ferritic, martensitic and precipitation hardening steels
- Identify and illustrate two commonly used materials-based strategies to prevent corrosion in metals
- Explain the role of Si and Al as alloying elements in the corrosion of alloys
- Enumerate three advantages of austenitic stainless steels and two disadvantages
- Clarify the sensitization phenomena, and specify which steels are susceptible to